



What is Amazon EC2?

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 eliminates your need to invest in hardware up front, so you can develop and deploy applications faster. You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.



Amazon
EC2

Agenda

- Features of Amazon EC2
- Related services
- Accessing Amazon EC2
- Pricing for Amazon EC2
- Instances and AMIs
- Getting started with Amazon EC2 Linux instances

Features of Amazon EC2

Amazon EC2 provides the following features:

- Virtual computing environments, known as instances
- Preconfigured templates for your instances, known as Amazon Machine Images (AMIs), that package the bits you need for your server (including the operating system and additional software)
- Various configurations of CPU, memory, storage, and networking capacity for your instances, known as instance types
- Secure login information for your instances using key pairs (AWS stores the public key, and you store the private key in a secure place)
- Storage volumes for temporary data that's deleted when you stop, hibernate, or terminate your instance, known as instance store volumes

Features of Amazon EC2

Amazon EC2 provides the following features:

- Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS), known as Amazon EBS volumes
- Multiple physical locations for your resources, such as instances and Amazon EBS volumes, known as Regions and Availability Zones
- A firewall that enables you to specify the protocols, ports, and source IP ranges that can reach your instances using security groups
- Static IPv4 addresses for dynamic cloud computing, known as Elastic IP addresses
- Metadata, known as tags, that you can create and assign to your Amazon EC2 resources
- Virtual networks you can create that are logically isolated from the rest of the AWS Cloud, and that you can optionally connect to your own network, known as virtual private clouds (VPCs)

Related services

You can provision Amazon EC2 resources, such as instances and volumes, directly using Amazon EC2. You can also provision Amazon EC2 resources using other services in AWS. For more information, see the following documentation:

- [Amazon EC2 Auto Scaling User Guide](#)
- [AWS CloudFormation User Guide](#)
- [AWS Elastic Beanstalk Developer Guide](#)
- [AWS OpsWorks User Guide](#)

To automatically distribute incoming application traffic across multiple instances, use Elastic Load Balancing.

Related services

To get a managed relational database in the cloud, use Amazon Relational Database Service (Amazon RDS) to launch a database instance. Although you can set up a database on an EC2 instance, Amazon RDS offers the advantage of handling your database management tasks, such as patching the software, backing up, and storing the backups. For more information, see the [Amazon Relational Database Service Developer Guide](#).

To make it easier to manage Docker containers on a cluster of EC2 instances, use Amazon Elastic Container Service (Amazon ECS). For more information, see the [Amazon Elastic Container Service Developer Guide](#) or the [Amazon Elastic Container Service User Guide for AWS Fargate](#).

To monitor basic statistics for your instances and Amazon EBS volumes, use Amazon CloudWatch.

To detect potentially authorized or malicious use of your EC2 instances, use Amazon GuardDuty.

Accessing Amazon EC2

Amazon EC2 provides a web-based user interface, the Amazon EC2 console. If you've signed up for an AWS account, you can access the Amazon EC2 console by signing into the AWS Management Console and selecting EC2 from the console home page.

If you prefer to use a command line interface, you have the following options:

AWS Command Line Interface (CLI)

Provides commands for a broad set of AWS products, and is supported on Windows, Mac, and Linux.

AWS Tools for Windows PowerShell

Provides commands for a broad set of AWS products for those who script in the PowerShell environment.

Accessing Amazon EC2

Amazon EC2 supports creating resources using AWS CloudFormation. You create a template, in JSON or YAML, that describes your AWS resources, and AWS CloudFormation provisions and configures those resources for you. You can reuse your CloudFormation templates to provision the same resources multiple times, whether in the same Region and account or in multiple Regions and accounts.

Amazon EC2 provides a Query API. These requests are HTTP or HTTPS requests that use the HTTP verbs GET or POST and a Query parameter named Action.

If you prefer to build applications using language-specific APIs instead of submitting a request over HTTP or HTTPS, AWS provides libraries, sample code, tutorials, and other resources for software developers. These libraries provide basic functions that automate tasks such as cryptographically signing your requests, retrying requests, and handling error responses, making it is easier for you to get started.

Pricing for Amazon EC2

When you sign up for AWS, you can get started with Amazon EC2 for free using the AWS Free Tier. Amazon EC2 provides the following purchasing options for instances:

On-Demand Instances

Pay for the instances that you use by the second, with no long-term commitments or upfront payments.

Savings Plans

You can reduce your Amazon EC2 costs by making a commitment to a consistent amount of usage, in USD per hour, for a term of 1 or 3 years.

Reserved Instances

You can reduce your Amazon EC2 costs by making a commitment to a specific instance configuration, including instance type and Region, for a term of 1 or 3 years.

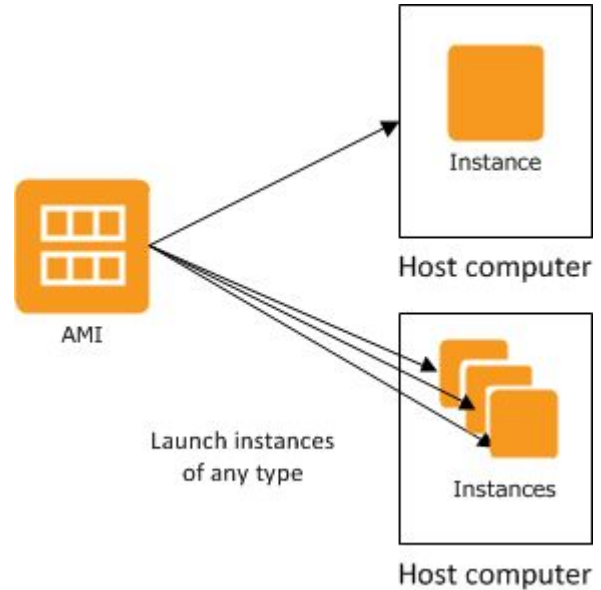
Spot Instances

Request unused EC2 instances, which can reduce your Amazon EC2 costs significantly.

Instances and AMIs

An Amazon Machine Image (AMI) is a template that contains a software configuration (for example, an operating system, an application server, and applications). From an AMI, you launch an instance, which is a copy of the AMI running as a virtual server in the cloud. You can launch multiple instances of an AMI, as shown in the following figure.

Your instances keep running until you stop, hibernate, or terminate them, or until they fail. If an instance fails, you can launch a new one from the AMI.



Instances

An instance is a virtual server in the cloud. Its configuration at launch is a copy of the AMI that you specified when you launched the instance.

You can launch different types of instances from a single AMI. An instance type essentially determines the hardware of the host computer used for your instance. Each instance type offers different compute and memory capabilities. Select an instance type based on the amount of memory and computing power that you need for the application or software that you plan to run on the instance.

After you launch an instance, it looks like a traditional host, and you can interact with it as you would any computer. You have complete control of your instances; you can use `sudo` to run commands that require root privileges.

Your AWS account has a limit on the number of instances that you can have running.

Instances: Storage for your instance

The root device for your instance contains the image used to boot the instance.

Your instance may include local storage volumes, known as instance store volumes, which you can configure at launch time with block device mapping. After these volumes have been added to and mapped on your instance, they are available for you to mount and use. If your instance fails, or if your instance is stopped or terminated, the data on these volumes is lost; therefore, these volumes are best used for temporary data. To keep important data safe, you should use a replication strategy across multiple instances, or store your persistent data in Amazon S3 or Amazon EBS volumes.

Instances: Security best practices

- Use AWS Identity and Access Management (IAM) to control access to your AWS resources, including your instances. You can create IAM users and groups under your AWS account, assign security credentials to each, and control the access that each has to resources and services in AWS.
- Restrict access by only allowing trusted hosts or networks to access ports on your instance. For example, you can restrict SSH access by restricting incoming traffic on port 22.
- Review the rules in your security groups regularly, and ensure that you apply the principle of least privilege—only open up permissions that you require. You can also create different security groups to deal with instances that have different security requirements. Consider creating a bastion security group that allows external logins, and keep the remainder of your instances in a group that does not allow external logins.
- Disable password-based logins for instances launched from your AMI. Passwords can be found or cracked, and are a security risk.

Instances: Stopping and terminating instances

You can stop or terminate a running instance at any time.

Stopping an instance

When an instance is stopped, the instance performs a normal shutdown, and then transitions to a stopped state. All of its Amazon EBS volumes remain attached, and you can start the instance again at a later time.

You are not charged for additional instance usage while the instance is in a stopped state. A minimum of one minute is charged for every transition from a stopped state to a running state. If the instance type was changed while the instance was stopped, you will be charged the rate for the new instance type after the instance is started. All of the associated Amazon EBS usage of your instance, including root device usage, is billed using typical Amazon EBS prices.

When an instance is in a stopped state, you can attach or detach Amazon EBS volumes. You can also create an AMI from the instance, and you can change the kernel, RAM disk, and instance type.

Instances: Stopping and terminating instances

Terminating an instance

When an instance is terminated, the instance performs a normal shutdown. The root device volume is deleted by default, but any attached Amazon EBS volumes are preserved by default, determined by each volume's `deleteOnTermination` attribute setting. The instance itself is also deleted, and you can't start the instance again at a later time.

To prevent accidental termination, you can disable instance termination. If you do so, ensure that the `disableApiTermination` attribute is set to `true` for the instance. To control the behavior of an instance shutdown, such as `shutdown -h` in Linux or `shutdown` in Windows, set the `instanceInitiatedShutdownBehavior` instance attribute to `stop` or `terminate` as desired. Instances with Amazon EBS volumes for the root device default to `stop`, and instances with instance-store root devices are always terminated as the result of an instance shutdown.

AMIs

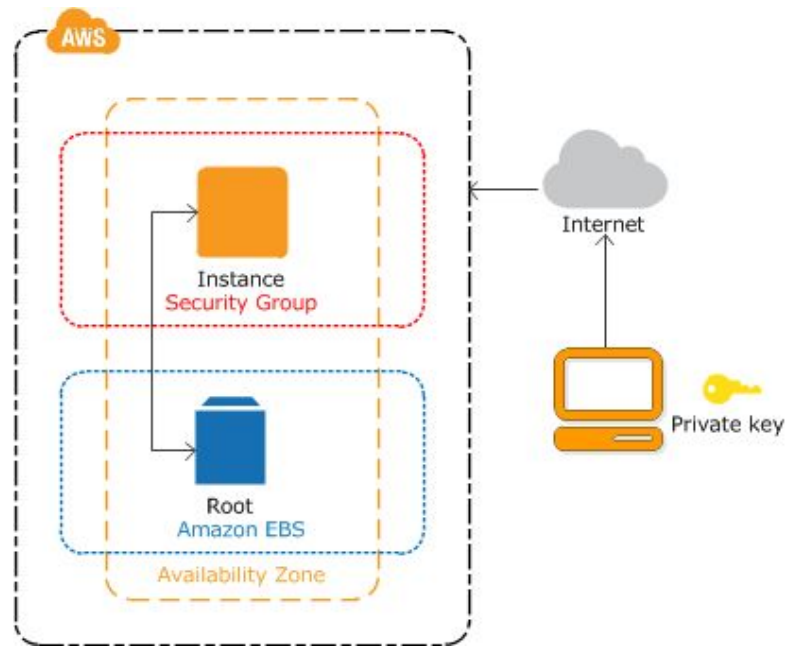
Amazon Web Services (AWS) publishes many Amazon Machine Images (AMIs) that contain common software configurations for public use. In addition, members of the AWS developer community have published their own custom AMIs. You can also create your own custom AMI or AMIs; doing so enables you to quickly and easily start new instances that have everything you need. For example, if your application is a website or a web service, your AMI could include a web server, the associated static content, and the code for the dynamic pages. As a result, after you launch an instance from this AMI, your web server starts, and your application is ready to accept requests.

All AMIs are categorized as either backed by Amazon EBS, which means that the root device for an instance launched from the AMI is an Amazon EBS volume, or backed by instance store, which means that the root device for an instance launched from the AMI is an instance store volume created from a template stored in Amazon S3.

The description of an AMI indicates the type of root device (either ebs or instance store). This is important because there are significant differences in what you can do with each type of AMI. You can deregister an AMI when you have finished using it. After you deregister an AMI, you can't use it to launch new instances. Existing instances launched from the AMI are not affected. Therefore, if you are also finished with the instances launched from these AMIs, you should terminate them.

Getting started with Amazon EC2 Linux instances

The instance is an Amazon EBS-backed instance (meaning that the root volume is an EBS volume). You can either specify the Availability Zone in which your instance runs, or let Amazon EC2 select an Availability Zone for you. When you launch your instance, you secure it by specifying a key pair and security group. When you connect to your instance, you must specify the private key of the key pair that you specified when launching your instance.



Tasks

We will perform the following tasks:

1. Launch an instance
2. Connect to Your Instance
3. Clean up your instance

Step 1: Launch an instance

You can launch a Linux instance using the AWS Management Console as described in the following procedure. This tutorial is intended to help you launch your first instance quickly, so it doesn't cover all possible options.

To launch an instance

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. From the console dashboard, choose **Launch Instance**.
3. The **Choose an Amazon Machine Image (AMI)** page displays a list of basic configurations, called Amazon Machine Images (AMIs), that serve as templates for your instance. Select an HVM version of Amazon Linux 2. Notice that these AMIs are marked "Free tier eligible."
4. On the **Choose an Instance Type** page, you can select the hardware configuration of your instance. Select the t2.micro instance type, which is selected by default. The t2.micro instance type is eligible for the free tier. In Regions where t2.micro is unavailable, you can use a t3.micro instance under the free tier.

4. On the **Choose an Instance Type** page, you can select the hardware configuration of your instance. Select the t2.micro instance type, which is selected by default. The t2.micro instance type is eligible for the free tier. In Regions where t2.micro is unavailable, you can use a t3.micro instance under the free tier.
5. Choose **Review and Launch** to let the wizard complete the other configuration settings for you.
6. On the **Review Instance Launch** page, under **Security Groups**, you'll see that the wizard created and selected a security group for you. You can use this security group, or alternatively you can select the security group that you created when getting set up using the following steps:
 - a. Choose **Edit security groups**.
 - b. On the **Configure Security Group** page, ensure that **Select an existing security group** is selected.
 - c. Select your security group from the list of existing security groups, and then choose **Review and Launch**.
7. On the Review Instance Launch page, choose Launch.
8. When prompted for a key pair, select Choose an existing key pair, then select the key pair that you created when getting set up. When you are ready, select the acknowledgement check box, and then choose **Launch Instances**.

Step 1: Launch an instance

9. A confirmation page lets you know that your instance is launching. Choose **View Instances** to close the confirmation page and return to the console.
10. On the **Instances** screen, you can view the status of the launch. It takes a short time for an instance to launch. When you launch an instance, its initial state is pending. After the instance starts, its state changes to running and it receives a public DNS name. (If the **Public DNS (IPv4)** column is hidden, choose **Show/Hide Columns** (the gear-shaped icon) in the top right corner of the page and then select **Public DNS (IPv4)**.)
11. It can take a few minutes for the instance to be ready so that you can connect to it. Check that your instance has passed its status checks; you can view this information in the **Status Checks** column.

Step 2: Connect to your instance

Step 3: Clean up your instance

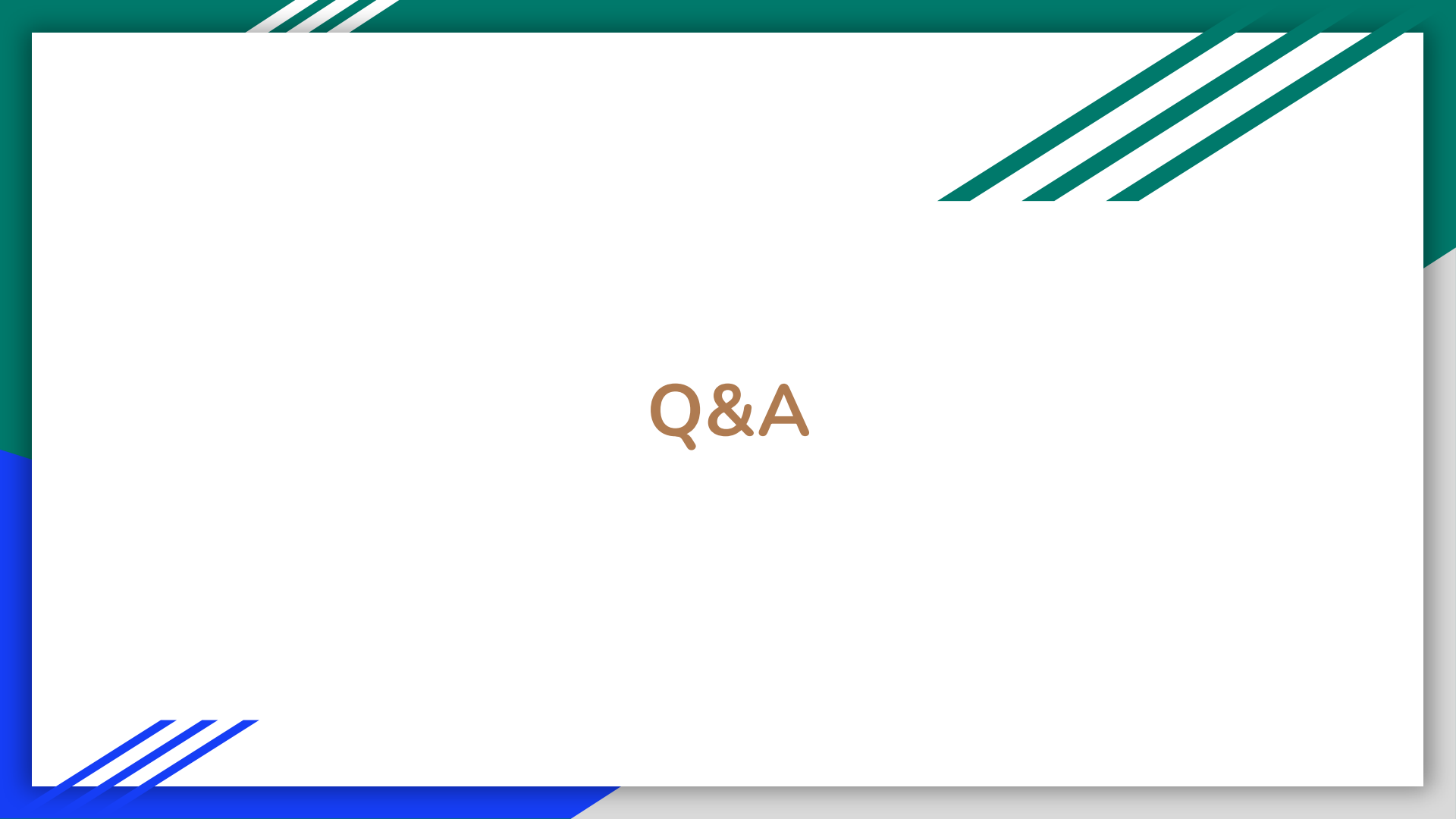
After you've finished with the instance that you created for this tutorial, you should clean up by terminating the instance.

If you launched an instance that is not within the AWS Free Tier, you'll stop incurring charges for that instance as soon as the instance status changes to shutting down or terminated. If you'd like to keep your instance for later, but not incur charges, you can stop the instance now and then start it again later.

To terminate your instance

1. In the navigation pane, choose **Instances**. In the list of instances, select the instance.
2. Choose **Instance state, Terminate instance**.
3. Choose **Terminate** when prompted for confirmation.

Amazon EC2 shuts down and terminates your instance. After your instance is terminated, it remains visible on the console for a short while, and then the entry is automatically deleted. You cannot remove the terminated instance from the console display yourself.



Q&A